# Franky Lian

lianfranky.10@gmail.com | (917) 306-0978 | Columbus, OH | www.linkedin.com/in/franky-lian

## EDUCATION

**Ohio State University - Columbus** **Expected Graduation: December 2027**
Bachelor of Science in Mechanical Engineering GPA: 3.95/4.00

## WORK EXPERIENCE

**Advanced Casting Research Center (OSU)** **Columbus, OH**
**Undergraduate Research Assistant** **March 2026 – Present**
• Conducted MAGMASOFT simulations across 11+ materials to evaluate solidification, fluidity, and porosity formation.
• Collected and analyzed experimental die-casting data to validate simulations and calibrate machine-learning models for casting defect prediction.
• Prepared metal samples for electron microscopy and tensile testing to characterize material and casting performance.

**GE Appliances, a Haier Company** **Louisville, KY**
**Mechanical Engineering Intern – Quality** **May 2026 – August 2026**
• Conducted dimensional analysis and tolerance stack-up studies on dishwasher components, providing engineering data that supported a cost reduction project with a projected $1M–$3M savings opportunity.
• Performed hardware evaluations, testing, and engineering validation to support quality and product performance improvements.
• Supported root cause investigation of component assembly failures using PolyWorks scan-to-CAD analysis, contributing to drawing revisions that reduced failure rates from 3% to 0.5% and saved approximately $150K.

**TotalSim LLC** **Dublin, OH**
**Computational Fluid Dynamics Engineering Intern** **February 2026 – May 2026**
• Created Siemens NX parameterizable models of impellers, wind tunnels, data center coolers, and engine ducts with 1%–2% deviation from legacy benchmarks.
• Cleaned complex CAD and scan-based geometries and generated CFD meshes using ANSA and STAR-CCM+ for aerodynamic trade studies.

**GE Appliances, a Haier Company** **Louisville, KY**
**Mechanical Engineering Intern – Design** **August 2025 – January 2026**
• Developed GD&T schemes (ASME Y14.5-2009) and technical specifications for 3 suspension subsystems.
• Validated suspension performance improvements against legacy designs through testing to support design decisions.
• Released engineering drawings on Creo and managed part configurations in Windchill PLM for new product development.
• Prototyped proofs-of-concepts for product challenges, contributing to 4 pending patent applications.

**TotalSim LLC** **Dublin, OH**
**Computational Fluid Dynamics Engineering Intern** **May 2025 – August 2025**
• Leveraged Siemens HEEDS and Java-based automation to enable repeatable CFD design studies, identifying impeller geometry changes that improved hydraulic efficiency by up to 5%.
• Performed CFD geometry preparation, case setup, and post-processing in STAR-CCM+ for UAV, thermal management, and rotating machinery applications.

## PROJECTS

**Liquid Rocket Systems Sub-team, Buckeye Space Launch Initiative** **August 2024 – Present**
• Developed a parametric Venturi nozzle in Onshape with configurable geometries to encompass different fasteners and CFD design requirements.
• Designed piston and O-ring dimensions for the rocket's aft closure, ensuring proper sealing and fit.

**Aerodynamics Sub-team, Buckeye Solar Racing** **September 2023 – March 2025**
• Constructed carbon-fiber composite structures such as the aeroshell, baffles, and other aerodynamic components to achieve lightweight and durable builds.

**Piston and Cylinder of Air Motor, ME2900 Machining Project** **January 2025 – February 2025**
• Machined aluminum components to 0.001-inch tolerance using CNC and manual lathe.
• Assembled and physically tested the air motor, achieving rotational speeds of 400+ RPM.

**3D Printed Centrifugal Compressor, Personal Project** **October 2024 – January 2025**
• Modeled a centrifugal compressor in SOLIDWORKS and analyzed airflow using ANSYS Fluent.
• Validated performance using an anemometer, showing experimental output was ±20% of the 24.55 CFM prediction.

## TECHNICAL SKILLS

**CAD & Design:** Creo, Siemens NX, SOLIDWORKS (CSWP), Onshape, GD&T, Schematics
**Simulation & Analysis:** Star-CCM+, ANSYS, Simcenter 3D, MagmaSOFT, Siemens HEEDS, PolyWorks, 3D-Scanning, Metrology
**Programming:** Python, MATLAB, Simulink, C++, Java, Linux
**Manufacturing:** CNC Machining, Manual Lathe, Milling, Composite Fabrication, Root Cause Analysis, PFMEA